

Trimly – App Data Flow

Trimly Development Team

1 How Trimly Works

Trimly is a salon booking and management app. Customers can find salons, select services, book appointments, and pay a deposit. Barbers can manage appointments, while salon owners can monitor their salon.

The app has three main users:

- Customer
- Barber
- Salon Owner

The React Native app sends requests to the backend. The backend processes requests and stores information in MongoDB. External services handle payments, OTPs, and notifications.

2 Complete App Data Flow

This diagram shows how the main parts of Trimly communicate.

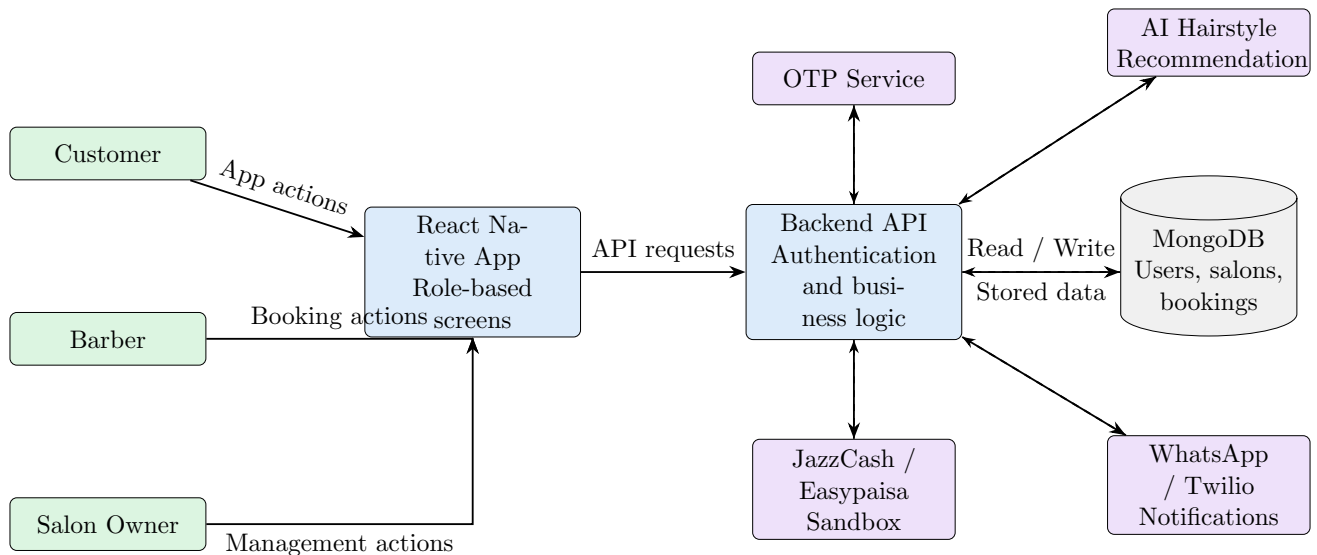


Figure 1: Complete Trimly data flow

What happens in this flow?

1. A user performs an action in the mobile app.
2. The app sends a request to the backend.

3. The backend checks the user's role and request.
4. The backend reads or updates the required data in MongoDB.
5. If needed, the backend communicates with an external service.
6. The backend sends the result back to the app.
7. The app displays the updated information to the user.

3 Customer Data Flow

The customer uses Trimly to find a salon, choose services, book an appointment, and pay the required deposit.

3.1 Customer Registration and Login

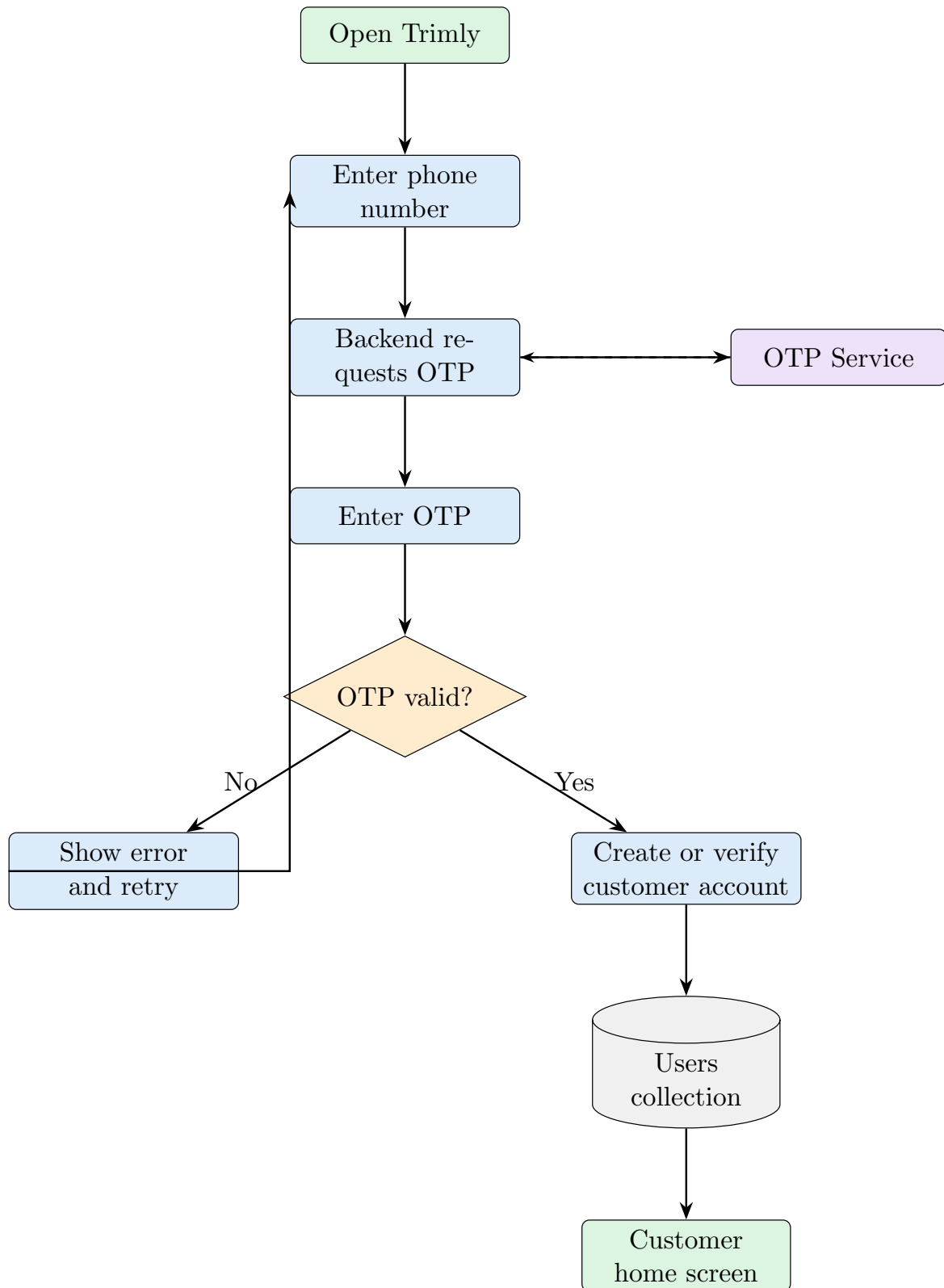
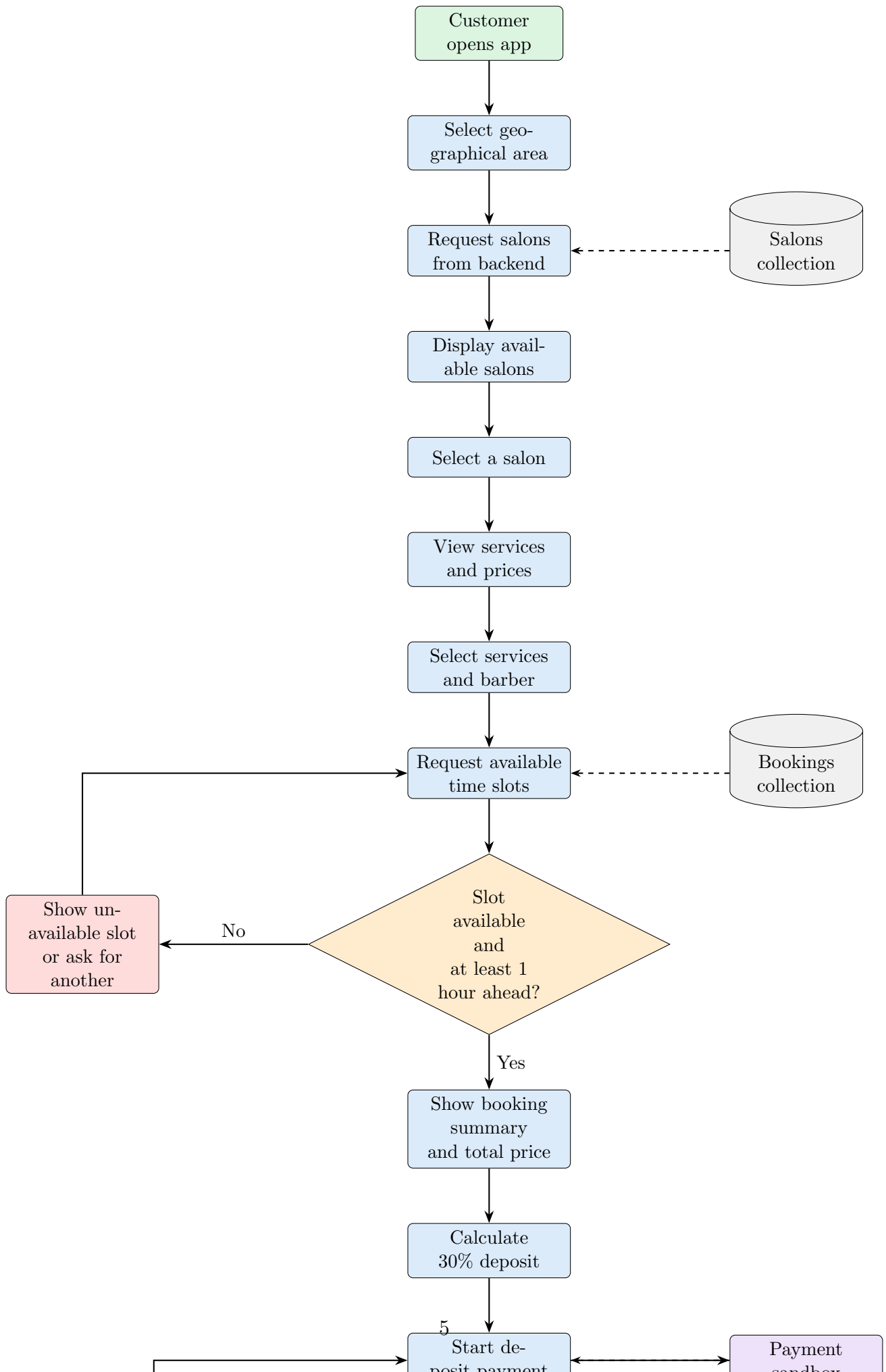


Figure 2: Customer login flow

3.2 Salon Search and Appointment Booking



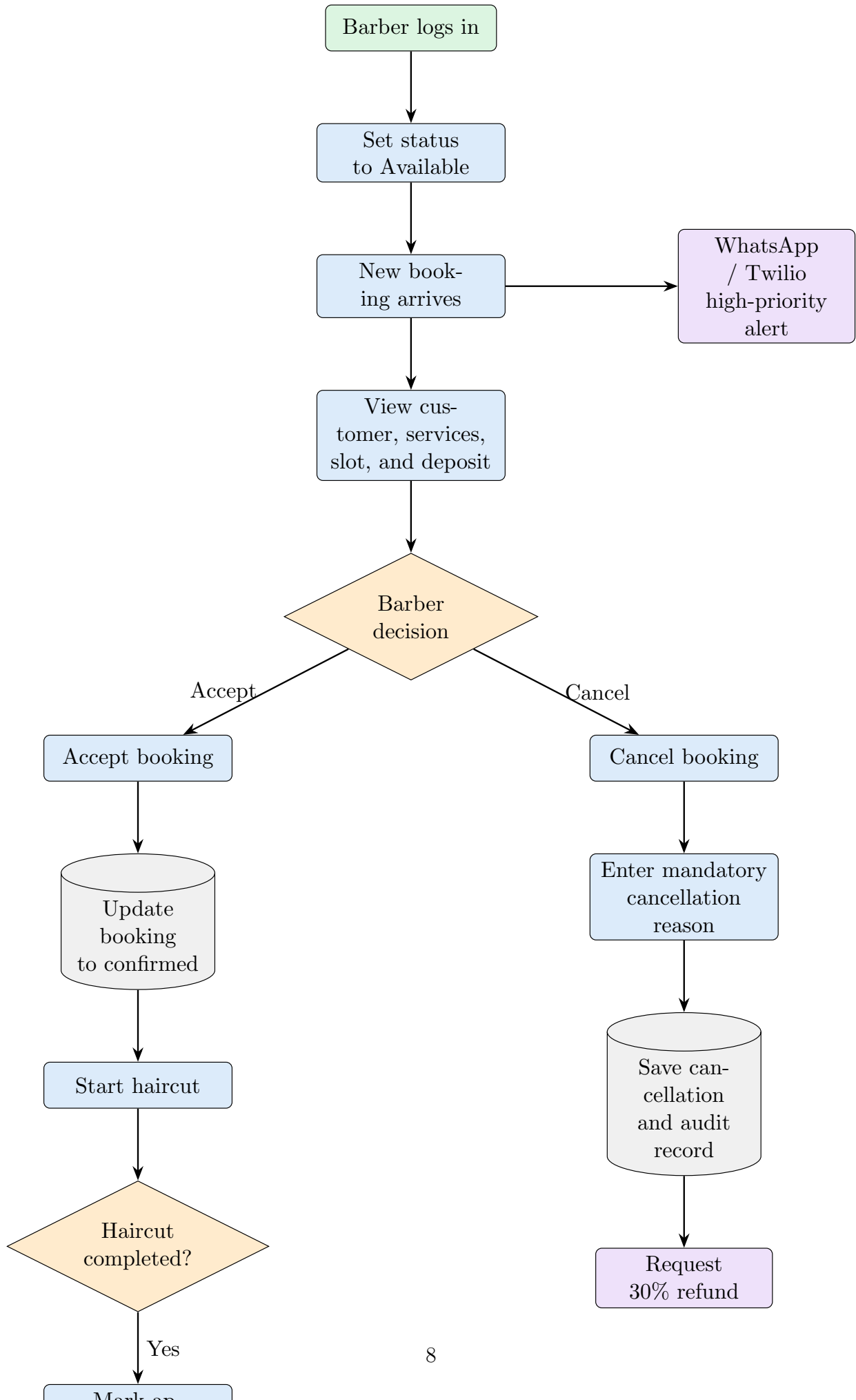
Booking process explained

1. The customer selects an area, such as North Nazimabad.
2. The backend returns salons available in that area.
3. The customer opens a salon and views its services.
4. The customer selects a barber, services, and a time slot.
5. The backend checks whether the slot is free and whether it is at least one hour in advance.
6. The app displays the booking summary and total price.
7. The backend calculates the 30% deposit.
8. The customer pays through the payment sandbox.
9. After payment confirmation, the backend creates the booking and stores the payment reference.
10. The selected barber receives a booking alert.

4 Barber Data Flow

The barber receives booking requests and decides whether to accept or cancel them. The barber can also start and complete a haircut.

4.1 Booking Decision Flow



Barber process explained

1. The barber logs in and sets their status to Available.
2. When a new booking arrives, the backend sends an alert.
3. The barber checks the booking details.
4. The barber has a 10-minute decision window.
5. If the barber accepts, the booking is confirmed.
6. If the barber cancels, they must enter a reason.
7. The backend saves the cancellation reason and adds an audit record.
8. The backend requests a refund of the 30% deposit.
9. Once the appointment is confirmed, the barber can start the haircut.
10. After completing the haircut, the barber marks it as completed.

If the barber does not cancel within the 10-minute decision window, the system can automatically confirm the booking according to the defined booking rules. The backend must prevent a late cancellation from overriding an already confirmed booking.

5 Salon Owner Data Flow

The salon owner manages the salon, staff, chairs, and services. The owner can also view reports and cancellation records.

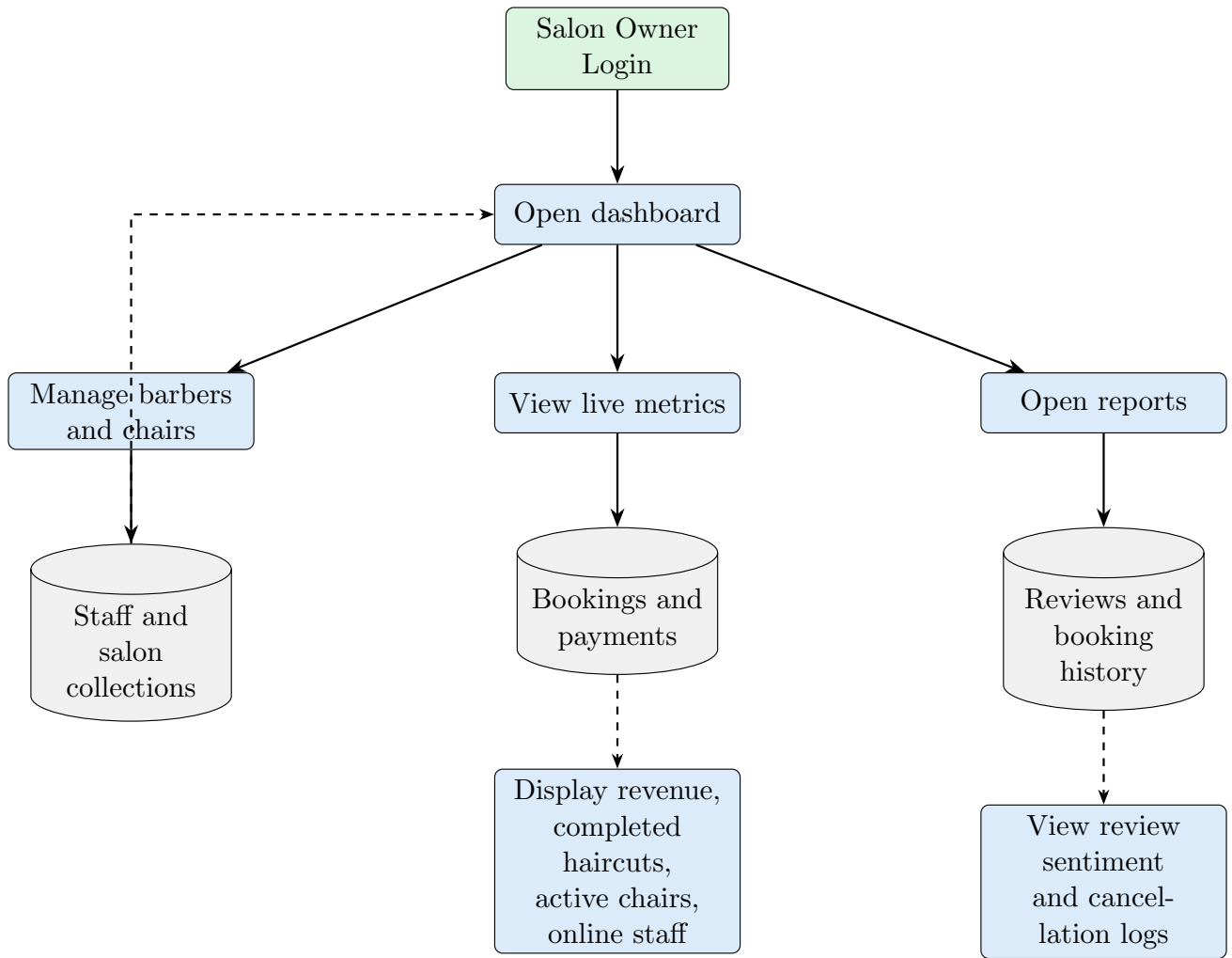


Figure 5: Salon owner dashboard flow

Owner process explained

1. The owner logs in and opens the dashboard.
2. The backend verifies the owner's salon access.
3. The owner can invite barbers and assign chairs.
4. The owner can update the salon profile and services.
5. The backend retrieves booking and payment data.
6. The dashboard displays revenue, completed haircuts, active chairs, and online staff.
7. The owner can review cancellation records and reasons.
8. The owner can view weekly revenue and staff utilization reports.
9. Negative review sentiment can trigger an alert.

6 Payment Data Flow

Customers pay 30% of the total amount before the appointment. The remaining 70% is paid at the salon after the service.

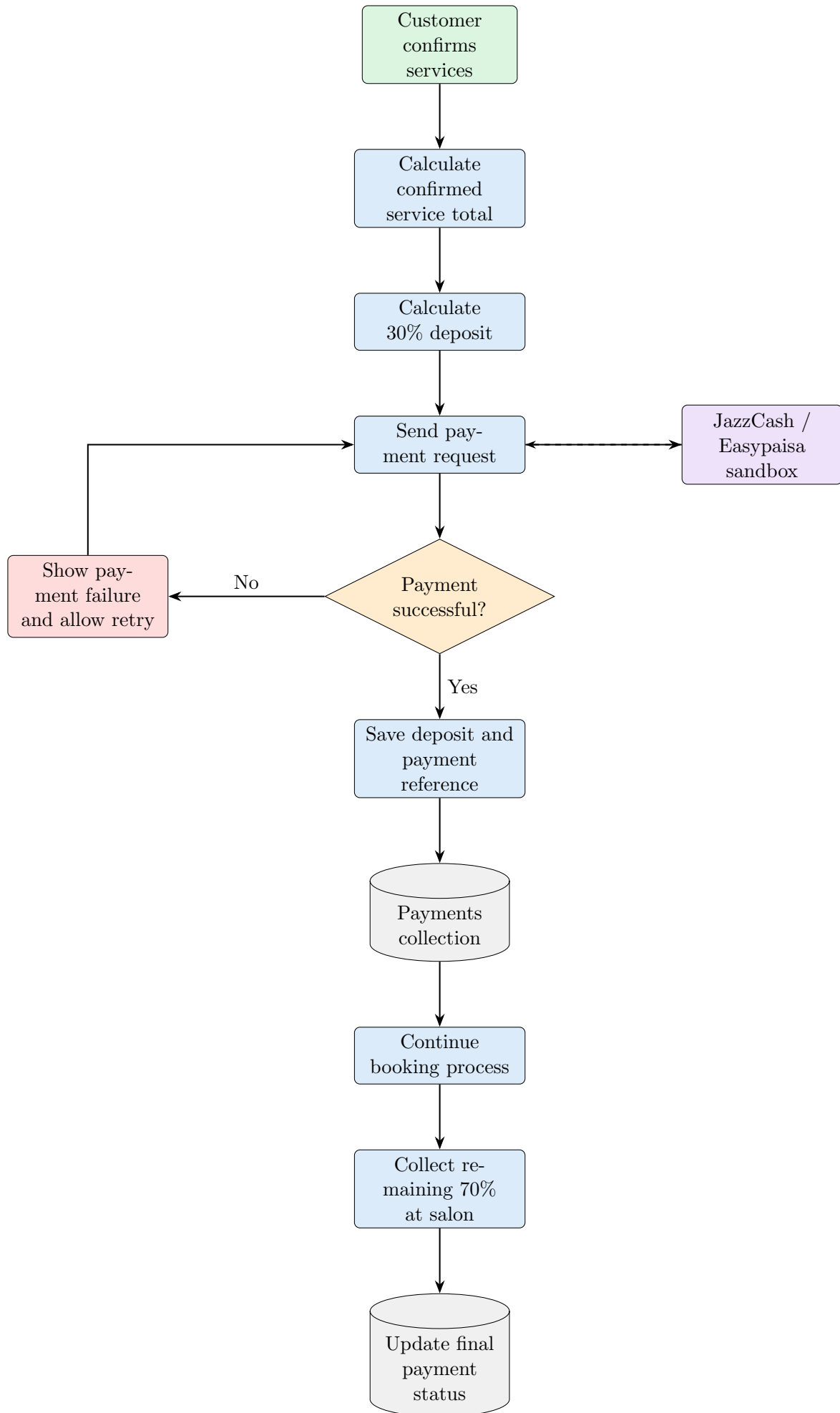


Figure 6: Customer payment flow

For example, if the confirmed services cost Rs. 2,000:

- Total amount: Rs. 2,000
- Advance deposit (30%): Rs. 600
- Remaining amount (70%): Rs. 1,400

The backend should store payment status separately from the booking status. A refund request should remain pending until the payment provider confirms its result.

7 AI Hairstyle Recommendation Flow

The AI hairstyle feature is optional and paid. It becomes available after the customer has paid the required deposit and purchased or unlocked the feature.

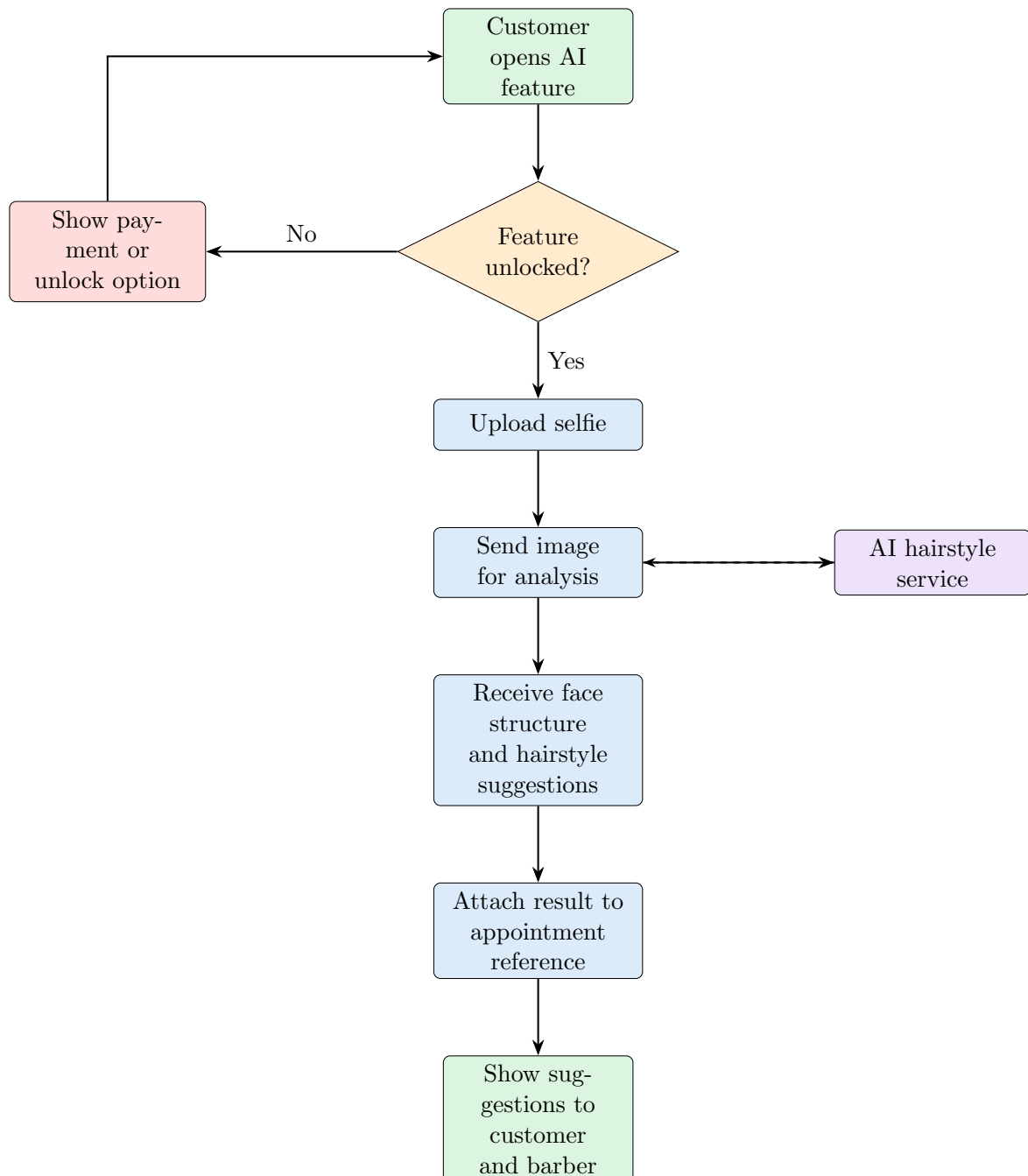


Figure 7: AI hairstyle recommendation flow

AI process explained

1. The customer opens the AI hairstyle feature.
2. The backend checks whether the feature is unlocked.
3. If it is unlocked, the customer uploads a selfie.
4. The backend sends the image to the AI service.
5. The AI service returns face-structure analysis and hairstyle suggestions.
6. The app displays the suggestions to the customer.
7. The result can be attached to the appointment for the barber.

8 Appointment Reminder Flow

The system sends a reminder 30 minutes before the appointment. The reminder is scheduled using the appointment time, regardless of whether the customer uses the AI feature.

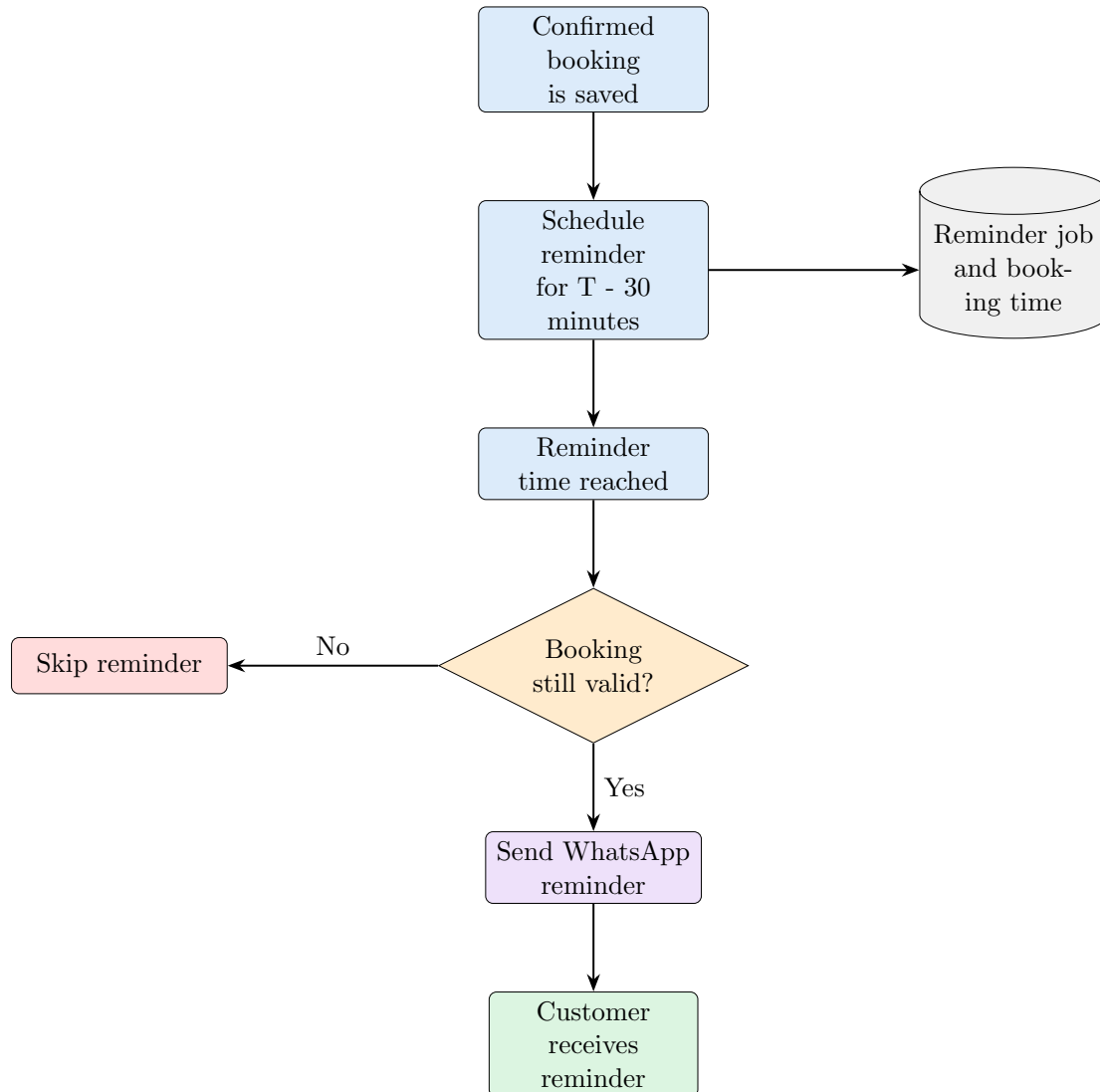


Figure 8: Appointment reminder flow

9 Review and Loyalty Points Flow

After an appointment is completed, the customer can submit a review. The system can then update loyalty points and analyze the review.

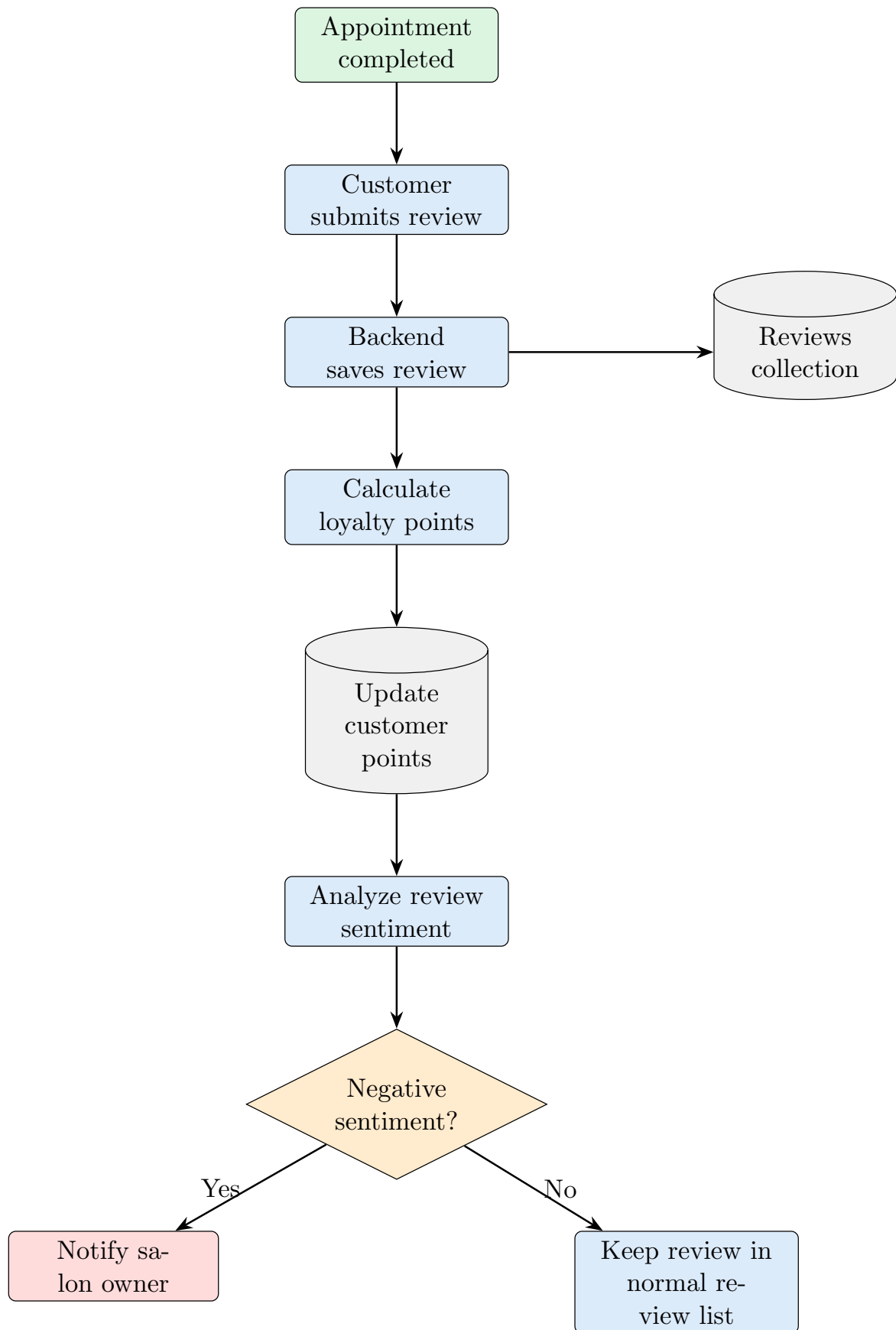


Figure 9: Review and loyalty flow

10 Main MongoDB Collections

These are the main collections needed to store the app's data.

- **users:** Customer, barber, and salon owner accounts.
- **salons:** Salon name, location, services, and profile.
- **barbers:** Barber profile, availability, and salon assignment.
- **services:** Service names, prices, and estimated duration.
- **bookings:** Customer, barber, salon, service, time, and status.
- **payments:** Deposits, remaining payments, and refunds.
- **reviews:** Customer ratings and comments.
- **loyaltyPoints:** Customer points and point history.
- **cancellationLogs:** Cancellation reason and related booking details.
- **notifications:** Reminder and booking notification records.
- **aiRecommendations:** AI results linked to a customer or appointment.

11 Important Booking Statuses

- **Pending:** The booking is waiting for the barber's decision.
- **Confirmed:** The booking has been accepted or automatically confirmed.
- **Cancelled:** The booking has been cancelled.
- **In Progress:** The barber has started the haircut.
- **Completed:** The haircut has been completed.

Payment statuses should be maintained separately:

- Pending
- Deposit Paid
- Fully Paid
- Refund Pending
- Refunded
- Failed

12 Important Rules

1. Customers must book at least 60 minutes in advance.
2. A barber has 10 minutes to accept or cancel a booking.
3. A cancellation requires a reason.
4. A cancellation should create an audit record.
5. The deposit is 30% of the confirmed service total.
6. The remaining 70% is paid at the salon.
7. The appointment reminder is scheduled for 30 minutes before the appointment.
8. AI hairstyle recommendations are an optional paid feature.
9. Only completed appointments should qualify for the review and loyalty process.
10. The backend must check booking availability before confirming a slot.

13 Final Overview

The main data flow of Trimly is:

1. The customer searches for a salon and books a service.
2. The backend validates the slot and processes the deposit.
3. The booking is sent to the selected barber.
4. The barber accepts or cancels the booking.
5. The system updates the booking and payment records.
6. The customer receives appointment reminders.
7. The barber completes the haircut and updates its status.
8. The customer pays the remaining amount and can submit a review.
9. The salon owner monitors bookings, staff, payments, and reports.

This document describes the main flow of data between users, the app, the backend, the database, and external services. It can be updated as the team finalizes the implementation.